Started on	Wednesday, 7 October 2026, 10:06 PM
State	Finished
Completed on	Wednesday, 7 October 2026, 10:15 PM
Time taken	8 mins 46 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>

int main(){
    int n;
    scanf("%d",&n);
    int a[n], dp[n];
    int i,j,max = 1;
    for(i = 0;i < n;i++){
        scanf("%d",&a[i]);
        dp[i] = 1;
    }
    for(i = 1;i < n;i++){
        for(j = 0; j < i;j++){
            if(a[j] <= a[i]&& dp[j] + 1 > dp[i]){
                dp[i] = dp[j] + 1;
            }
        }
```

```
        if(dp[i] > max)
            max = dp[i];

    }
    printf("%d",max);

    return 0;
}
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.